A

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ANNUAL EVALUATION 2024-2025

E-803

Class : 8

MATHEMATICS

Time : 1 ½ hours

Score : 40

Instructions

- There is a 'cool off' time of 15 minutes in addition to the writing time. Use this time to get familiar with questions and plan your answers.
- Read the instructions carefully before answering the questions.
- Keep in mind, the score and time while answering the questions. Give explanations wherever necessary.

Answer any 4 questions from 1 to 5. Each question carries 2 scores. **(4 x 2 = 8)**

1. In the figure,the line PR is divided into four equal parts.



   a) PQ : QR = _____

   b) If the length of PQ is 6 centimetres,what is the length of QR ?

2. a) Write $\frac{1}{25}$ as a fraction with denominator a power of 10.

   b) Write the decimal form of $\frac{1}{25}$

3. One angle of an isosceles triangle is 130^0. What are the other two angles ?

4. Five added to two times a number gives 21. What is the number?

5. Length of the diagonal of a square is 10 centimetres . Find the area of the square.

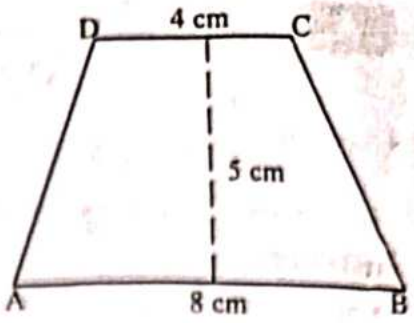
Answer any 4 questions from 6 to 11. Each question carries 3 scores. **(4 x 3 = 12)**

6. A person deposited 6000 rupees in a bank which compounds interest quarterly. Annual rate of interest is 12% .

   a) What is the quarterly rate of interest?
   (12%, 6%, 4%, 3%)

   b) How much would he get back after six months ?

7. In the figure, ABCD is a trapezium.
AB = 8 centimetres, DC = 4 centimetres.
The distance between the parallel sides is 5 centimetres.



a) What is the sum of parallel sides of the trapezium?

b) Find the area of the trapezium.

8. $x^2 - y^2 = (x + y)(x - y)$

a) If $x + y = 10, x - y = 8$,
then $x^2 - y^2 =$ _____

b) Using the given identity, find 51×49.

9. The weights of 30 children in a class are listed below.

28, 26, 33, 36, 31, 39, 34, 43, 37, 29,
46, 49, 47, 37, 26, 21, 44, 34, 43, 39,
32, 28, 24, 36, 48, 42, 32, 33, 43, 38.

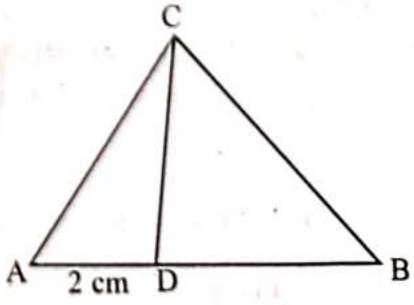
Using this data, complete the following frequncy table.

Weight (Kg)	Tally	Number of Children
20 - 25	.....	.....
25 - 30	.....	.....
30 - 35	.....	.....
35 - 40	.....	.....
40 - 45	.....	.....
45 - 50	.....	.....

10. a) Find the sum of angles of a 10 sided regular polygon.

b) What is the measure of one inner angle of this polygon ?

11. In the figure, AB = 5 centimetres, AD = 2 centimetres.



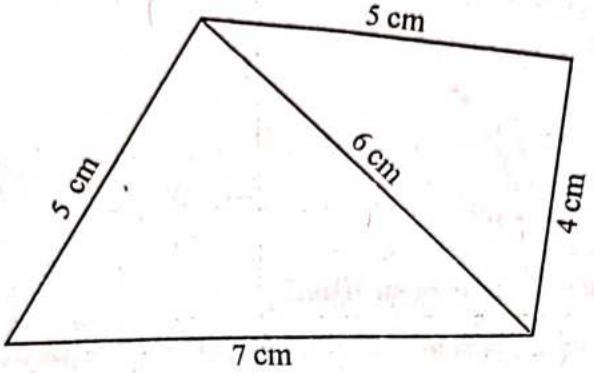
a) BD = _____

b) Write the ratio between the areas of triangle ADC and the triangle BDC.

c) If the area of triangle ABC is 35 square centimetres, find the area of triangle ADC.

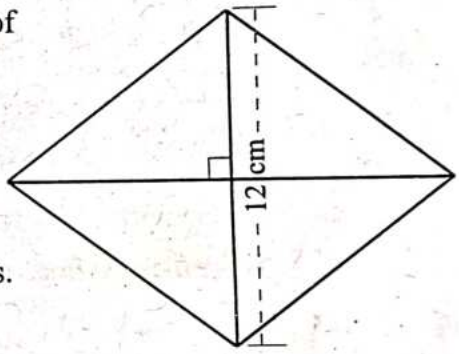
Answer any 5 questions from 12 to 18. Each question carries 4 scores. **(5 x 4 = 20)**

12. Draw the quadrilateral with the given measurements.



13. The angles of a triangle are in the ratio 2 : 3 : 4 .
    a) What is the sum of angles of a triangle ?
    b) Find the three angles of the given triangle.

14. The perimeter of a rhombus is 40 centimetres and length of one of its diagonal is 12 centimetres in the given figure.



    a) What is the length of one side of the rhombus?
    b) Find the length of the second diagonal of the rhombus.
    c) Find the area of this rhombus.

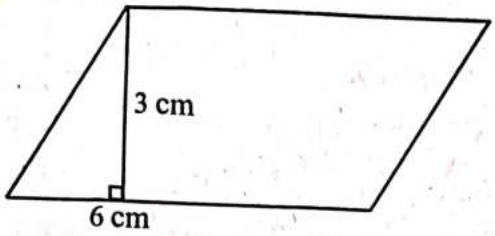
15. The table below shows the ages of 35 members in a Kudumbasree unit.

    Draw a Histogram.

Age	Number of members
20 - 30	4
30 - 40	7
40 - 50	14
50 - 60	5
60 - 70	3
70 - 80	2

16. Draw a triangle with sides 7 centimetres, 6 centimetres and 5 centimetres. Draw its circumcircle.

17.



a) Find the area of the parallelogram in the figure.

b) Draw a parallelogram with one side 7 centimetres and area 35 square centimetres.

18. Read and understand the mathematical concept in the given statements, write the answers to the following questions.

$1^2+2^2 = 3^2 - 2(1 \times 2)$

$2^2+3^2 = 5^2 - 2(2 \times 3)$

$3^2+4^2 = 7^2 - 2(3 \times 4)$

.....

.....

.....

a) Write the next line.

b) $7^2 + 8^2 = 15^2 - 2$ (_____)

c) Write the 20th line of this pattern.

d) $a^2 + b^2 =$ _____